The Compton Maintenance Hypothesis

Deriving MOND's Acceleration Scale from Quantum Phase Maintenance
and the Information-Theoretic Origin of the 2π Factor

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Abstract

We present a single physical postulate—that every massive particle maintains its existence through continuous quantum phase evolution at its Compton frequency, and that each complete phase cycle constitutes one irreversible bit of information requiring one Planck area of cosmological boundary surface—and show that this postulate, combined with the SFT master equation $H = \dot{S}/I$, produces a modified Friedmann equation $H^2 = 2G\rho/(3\pi)$. This differs from the standard Friedmann equation $H^2 = 8\pi G\rho/3$ by a factor of exactly $1/(4\pi^2)$, implying a natural acceleration threshold of $a_{SFT} = cH_0/(2\pi) \approx 1.05 \times 10^{-10}$ m/s^2. This value agrees with MOND's empirical acceleration constant $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 to within 13%, with zero free parameters. The 2π factor is not a geometric fit—it is the fundamental ratio between linear and angular frequency in quantum mechanics, identifying one bit of irreversible information with one complete cycle of quantum phase.

1. Introduction

The Modified Newtonian Dynamics (MOND) paradigm, introduced by Milgrom in 1983, successfully predicts the rotation curves of disk galaxies using a single acceleration threshold $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. Below this threshold, gravitational dynamics deviate from Newtonian predictions in a manner that eliminates the need for dark matter in galaxies. Despite four decades of empirical success across hundreds of galaxies, the physical origin of a_0 remains unexplained. Its approximate numerical coincidence with cH_0 has been noted by several authors, including Milgrom himself, but has been treated as either coincidental or suggestive without a derivation.

In a companion paper (SFT V7), we reported that a direct weak-field acceleration law with $a_{SFT} = cH_0/(2\pi)$ reduces rotation curve error by 63% across 137 SPARC galaxies with zero per-galaxy fitting. The reduced scale $cH_0/(2\pi)$ consistently outperformed both the literal scale cH_0 and other candidates. At the time, the factor of 2π was treated as an empirical finding without theoretical derivation.

This note provides that derivation.

2. The Compton Maintenance Postulate

In quantum mechanics, a massive particle is not a static object. Its quantum phase evolves at the Compton frequency:

$$\nu = mc^2 / h$$

where m is the particle's rest mass, c is the speed of light, and h is Planck's constant. This evolution is not optional—it is the definition of a massive quantum state. A particle that does not evolve its phase does not exist as a physical entity.

We propose the following postulate:

The Compton Maintenance Postulate. *Every massive particle maintains its existence through continuous quantum phase evolution. Each complete phase cycle (2π radians of quantum phase, corresponding to one linear Compton tick $\nu = mc^2/h$) constitutes one irreversible bit of information—one affirmation of the particle's continued reality. By the Bekenstein-Hawking process equation, each such bit demands exactly one Planck area ($\ell_P^2 = \hbar G/c^3$) of cosmological boundary surface to record it.*

This postulate makes three physical claims. First, existence is computational: a particle persists by continuously processing one bit per Compton cycle. Second, the relevant unit of information is one complete phase revolution (2π radians), not one radian of phase advance. Third, the geometric cost of each bit is exactly one Planck area, consistent with the Bekenstein-Hawking entropy formula read as a process equation rather than a state equation.

3. Derivation of the Modified Friedmann Equation

3.1 Total entropy production rate

Under the Compton maintenance postulate, the total information production rate of the universe is the sum of Compton frequencies over all massive particles. For a universe with total mass-energy M in the Hubble volume:

$$\dot{S}_total = Mc^2 / h$$

3.2 Information capacity of the horizon

The information storage capacity of the cosmological horizon is its Bekenstein-Hawking entropy. For a Hubble horizon of radius $R_H = c/H$:

$$I = A / (4\ell_P^2) = \pi c^2 / (\ell_P^2 H^2)$$

3.3 The master equation

The SFT master equation identifies the Hubble parameter as the fractional rate of information accumulation:

$$H = \dot{S}_total / I$$

Substituting:

$$H = [Mc^2/h] / [\pi c^2/(\ell_P^2 H^2)] = M\ell_P^2 H^2 / (\pi h)$$

3.4 Substituting mass and Planck area

We express the total mass as $M = \rho V = \rho \times (4\pi/3)(c/H)^3$, where ρ is the mean density, and the Planck area as $\ell_P^2 = \hbar G/c^3 = Gh/(2\pi c^3)$. Substituting both:

$$\mathbf{H = [\rho \times (4\pi/3)(c^3/H^3)] \times [Gh/(2\pi c^3)] \times [H^2/(\pi h)]}$$

The factors of c^3, h, and one factor of π cancel, yielding:

$$\mathbf{H = 2G\rho \,/\, (3\pi H)}$$

Multiplying both sides by H:

$$\mathbf{H^2 = 2G\rho \,/\, (3\pi)}$$

This is the SFT modified Friedmann equation, derived from the Compton maintenance postulate with no free parameters.

4. Origin of the 2π Factor

The standard Friedmann equation is $H^2 = 8\pi G\rho/3$. The ratio of the SFT equation to the standard equation is:

$$\mathbf{H^2_SFT \,/\, H^2_Friedmann \;=\; [2G\rho/(3\pi)] \,/\, [8\pi G\rho/3] \;=\; 1/(4\pi^2)}$$

Taking the square root:

$$\mathbf{H_SFT = H_Friedmann \,/\, (2\pi)}$$

The natural SFT acceleration scale is therefore:

$$\mathbf{a_SFT = cH_0 \,/\, (2\pi) \approx 1.05 \times 10^{-10}\ m/s^2}$$

The factor of 2π is not geometric. It is quantum mechanical. It arises because one irreversible bit of information corresponds to one complete cycle of quantum phase—2π radians—rather than one radian. This is the distinction between linear frequency $\nu = mc^2/h$ (which counts complete cycles) and angular frequency $\omega = mc^2/\hbar$ (which counts radians). If bits were counted per radian, the factor would vanish and $a_{SFT} = cH_0$ exactly. The SPARC data decisively prefer the linear frequency, confirming that the fundamental unit of irreversible quantum information is one complete phase revolution.

5. Comparison with Observation

Quantity	Value	Source
$a_SFT = cH_0/(2\pi)$	1.05×10^{-10} m/s^2	*This derivation (zero parameters)*
a_0 (MOND empirical)	1.2×10^{-10} m/s^2	*Milgrom 1983; McGaugh+ 2016*

Agreement	**13%**	**Zero free parameters**

In the SFT V7 weak-field test, the scale $cH_0/(2\pi)$ was applied to 137 quality-filtered SPARC galaxies using an exponential interpolation function with no per-galaxy fitting. This produced a 63% reduction in mean rotation curve error relative to baryons-only, outperforming both the literal scale cH_0 and the alternative $cH_0/6$. The data independently selected the 2π reduction before this derivation existed.

6. The $4\pi^2$ Structure of the Dark Energy Gap

The SFT Compton equation $H^2 = 2G\rho/(3\pi)$ differs from the standard Friedmann equation $H^2 = 8\pi G\rho/3$ by a factor of exactly $4\pi^2 \approx 39.5$. This factor is not approximate—it is an algebraic consequence of the Compton maintenance postulate. In this section we argue that this exact factor is not a failure of the derivation but a precise structural characterization of the dark energy problem.

6.1 The universal amplitude gap

Every framework that attempts to predict the cosmic expansion rate from known physics encounters the same problem. Standard matter-only Friedmann (without Λ) predicts an expansion rate approximately 44% below the observed H_0. The SFT Compton equation, stacked additively on top of standard gravity, closes only 2.5% of that gap. The remaining deficit is what ΛCDM attributes to a cosmological constant Λ—a term with no physical origin, no dynamical explanation, and no connection to any other scale in physics.

SFT does not solve this amplitude gap in the present work. But it does something no other framework has done: it gives the gap an **exact value** expressed in terms of fundamental constants. The ratio between the observed expansion rate and the Compton maintenance rate is exactly $(2\pi)^2 = 4\pi^2$. This is not a fitted number. It falls directly out of identifying one irreversible bit with one complete quantum phase cycle (2π radians) and one Planck area of boundary surface.

6.2 Comparison with ΛCDM

In ΛCDM, the dark energy gap is parameterized by $\Omega_\Lambda \approx 0.69$—an empirical number with no derivation. The only statement ΛCDM makes about why 69% of the universe's energy budget is dark energy is that it fits the data.

The Compton maintenance postulate provides a more specific statement: the expansion rate exceeds the information-maintenance rate by a factor whose square is $4\pi^2$, arising from the quantum-geometric relationship between Compton phase cycles and horizon boundary area. Whether or not SFT ultimately explains *why* the factor is $4\pi^2$, it has identified *what* the factor is—which is further than any framework has gone in expressing the dark energy gap as a derivable quantity rather than a free parameter.

6.3 A doorway, not a wall

Exact factors of π in physics invariably signal geometric structure. The appearance of $(2\pi)^2$ specifically suggests a relationship between linear and spherical geometry—the same kind of relationship that appears when mapping flat-space quantities onto curved boundaries. We note three structural observations:

First, the factor 4 in the Bekenstein-Hawking formula $S = A/(4\ell_P^2)$ means one bit occupies 4 Planck areas, not 1. If this factor is incorporated into the cost-per-tick, the deficit reduces from $4\pi^2$ to π^2.

Second, π^2 arises naturally in the mapping of uniform distributions onto spherical surfaces—precisely the geometry of projecting local Planck-scale information onto the spherical cosmological horizon.

Third, the Compton maintenance postulate already correctly produces the galactic acceleration scale from the same 2π factor. The deficit at cosmological scales is the *square* of the factor that works at galactic scales. This suggests the cosmological and galactic expressions may be related by a single geometric correction that applies once for the acceleration ratio and twice for the energy density ratio.

Resolving the $4\pi^2$ into a complete cosmological theory is the central task for future work. The present paper does not claim to have completed this resolution. It claims only that the Compton maintenance postulate has converted the dark energy problem from an unexplained empirical constant ($\Omega_\Lambda = 0.69$) into an exact geometric factor ($4\pi^2$) whose structure invites derivation rather than resisting it.

7. Discussion

The central claim of this note is narrow but significant: the factor of 2π that separates the observed galactic acceleration threshold from cH_0 is not a fitted parameter, a geometric projection, or a coincidence. It is the fundamental quantum mechanical conversion factor between linear and angular frequency, arising from the identification of one irreversible bit with one complete cycle of quantum phase.

This identification has a clear physical meaning. A particle's quantum phase must complete a full 2π revolution to return to its initial state. Only at that point has one complete unit of irreversible quantum computation been performed. Fractional phase advances (less than 2π) do not constitute completed operations in the information-theoretic sense. This is analogous to how a classical bit must fully transition from 0 to 1 to register a state change—partial transitions do not count.

The result connects three previously disconnected domains: quantum mechanics (Compton frequency), information theory (Bekenstein-Hawking entropy as a process equation), and galactic dynamics (MOND's acceleration threshold). Each connection is mediated by established physics. No new fields, particles, or modifications to general relativity are required.

What is required is a single physical assumption: that existence is computational, and that the universe charges for computation in units of boundary area. This assumption is consistent with the Bekenstein bound, the holographic principle, and the Landauer limit. The Compton maintenance

postulate adds specificity to these established principles by identifying the processing rate (Compton frequency) and the unit of irreversibility (one complete phase cycle).

8. Conclusion

From the single postulate that every massive particle pays a continuous information maintenance cost at its Compton frequency, with each complete phase cycle demanding one Planck area of cosmological boundary, we derive:

(1) A modified Friedmann equation $H^2 = 2G\rho/(3\pi)$, differing from the standard equation by exactly $1/(4\pi^2)$.

(2) A natural galactic acceleration threshold $a_{SFT} = cH_0/(2\pi) = 1.05 \times 10^{-10}$ m/s^2, agreeing with MOND's empirical $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 to 13% with zero free parameters.

(3) A physical identification of the 2π factor as the linear-to-angular frequency ratio in quantum mechanics, confirmed independently by the SPARC rotation curve data.

These results do not complete Solvency Field Theory, but they provide a concrete, falsifiable, and parameter-free prediction linking quantum phase evolution to cosmological and galactic dynamics. The Compton maintenance postulate offers a candidate answer to the question that has remained open since 1983: why does the universe have a preferred acceleration scale, and why does it coincide with cH_0?